



Bank Account Interest Calculator

Console and Swing GUI application in Java

By Sameer Shaikh | Roll No. 150096725026 | Cohort: Jeff Bezos

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Introduction, Problem & Objectives

Introduction

- Java program that takes account details, adds a bonus rate by account type, and calculates simple interest and maturity balance
- Runs as a console program (Scanner) and as a Swing window (JFrame)

Problem Statement

Calculating deposit interest by hand is repetitive, and the result changes with the account type. A program is needed that validates the details, applies the account-type rules, and shows interest and maturity amount.

Objectives

- 1 Take name, account number and balance, and validate them
- 2 Choose Savings, Fixed Deposit or Recurring Deposit
- 3 Apply account-specific bonus rates with if-else
- 4 Calculate simple interest and maturity balance
- 5 Show an account summary; handle wrong input with loops and try-catch
- 6 Provide a Swing GUI that reuses the same calculation methods

Solution, Features & Technologies

Proposed Solution

- Reads the account details, checks each value, and applies the rules for the chosen account type
- Shows interest earned and maturity balance in a formatted summary
- Calculation logic is written once in static methods; console and GUI both call it

Main Features

Input validation	Savings Account	Fixed Deposit
Recurring Deposit	Bonus-rate calculation	Simple interest
Maturity balance	Account summary	Console and GUI

Technologies

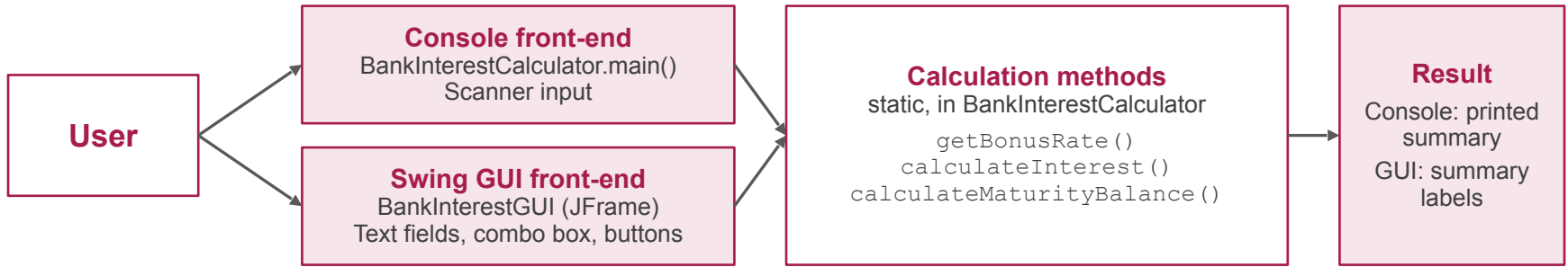
Language	Java, standard libraries only (no external jars, no build tool)
Console	java.util.Scanner, System.out, InputMismatchException
GUI	javax.swing (JFrame, JPanel, JLabel, JTextField, JComboBox, JButton, JOptionPane); java.awt (BorderLayout, GridLayout, Color, Font)
Needs	JDK on any OS; no database or network

Bonus rules (project assumptions)

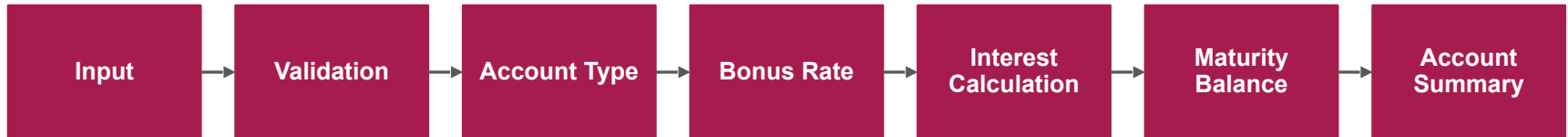
Account type	Condition	Bonus
Savings	balance $\geq$ 50000	0.5 %
Fixed Deposit	years $\geq$ 5 / years $\geq$ 2	1.0 % / 0.5 %
Recurring Deposit	years $\geq$ 3	0.25 %

System Architecture & Workflow

Architecture



Workflow



If a check fails: the console asks again (while loop); the GUI shows an error dialog and leaves the summary unchanged.

Implementation & Java Concepts

BankInterestCalculator static methods only

```
calculateInterest()  
calculateMaturityBalance()  
getBonusRate()  
displayAccountSummary()  
main() // console version
```

BankInterestGUI extends JFrame

```
calculateAndShow()  
readNumber()  
showError()  
clearAll(), resetSummary()  
main() // invokeLater
```

Formulas

Final Rate = Entered Rate + Bonus

Simple Interest = $(P \times R \times T) / 100$

Maturity Balance = Principal + Interest

Java concept	Where it is used
Static methods	All calculation methods
if-else / switch	getBonusRate() / account type in console
Loops	while loops repeat input until valid
Regex	Name and account number checks
Exceptions	InputMismatchException (console), NumberFormatException (GUI)
Array, generics	String[] accountTypes in JComboBox<String>
Inheritance	BankInterestGUI extends JFrame
Encapsulation	Private fields and methods in the GUI class
Event handling	Anonymous ActionListener on both buttons
Swing layouts	BorderLayout, GridLayout

GUI, Testing & Results

Bank Account Interest Calculator

Calculate Interest and Maturity Balance

1 Account Details

Account Holder Name

Account Number

Account Balance (Rs.)

Interest Rate (%)

Duration (Years)

2 Account Type

3 Calculate Interest **4 Clear**

5 Account Summary

Maturity Balance **6**

Rs. 1776589.08

Account Holder **ashish**

Account Number **134674**

Account Type **Savings Account**

Principal Amount **Rs. 746466.00**

Base Interest Rate **34.00 %**

Bonus Rate **0.50 %**

Final Interest Rate **34.50 %**

Duration **4.00 year(s)**

Interest Earned **Rs. 1030123.08**

- 1** Account Details
- 2** Account Type
- 3** Calculate Interest
- 4** Clear button
- 5** Account Summary
- 6** Maturity Balance

Test cases (expected values)

ID	Type	Balance / Rate % / Years	Bonus	Interest	Maturity
T01	Savings	100000 / 4 / 2	0.50	9000.00	109000.00
T03	Savings (boundary)	50000 / 4 / 1	0.50	2250.00	52250.00
T04	Fixed Deposit	200000 / 6 / 5	1.00	70000.00	270000.00
T06	Fixed Deposit	100000 / 6 / 1.5	0.00	9000.00	109000.00
T07	Recurring Deposit	60000 / 5 / 3	0.25	9450.00	69450.00
T09	Savings (zero)	0 / 5 / 2	0.00	0.00	0.00
T10	Savings (decimals)	75000.50 / 6.5 / 2.5	0.50	13125.09	88125.59

7 of the 10 documented cases shown; values are calculated from the formulas and bonus rules.

Invalid input (console): empty or non-letter name, non-digit account number, negative or non-numeric balance, rate outside 0–100, duration ≤ 0 , account type outside 1–3. Each was rejected and asked again.



Advantages, Limitations & Future Scope

Advantages

- Formulas and bonus rules written once, used by console and GUI
- Every input is checked; wrong data types do not crash the console program
- Boundary values (balance 50000, 2 years) tested
- Standard Java only, no extra setup

Limitations

- Simple interest only, no compound interest
- Recurring Deposit treated as one amount, no monthly instalments
- Bonus rules fixed in code; account type matched by text
- Validation repeated in console and GUI classes
- One account at a time, nothing saved, no JUnit tests

Future Scope

- Compound interest and monthly RD instalments
- enum for account types; one shared validation method
- Files or database storage; several accounts with ArrayList
- BigDecimal for money values; JUnit tests

Conclusion

- Calculates interest and maturity balance for three account types and validates input at every step
- Console and GUI share the same calculation methods
- Interest is simple interest and bonus rules are project assumptions, not real bank rates